

Salman Khan

AI Engineer

+92-3415981261 | salmank.official751@gmail.com | [linkedin.com/in/salman-khan-ai-software-engineer/](https://www.linkedin.com/in/salman-khan-ai-software-engineer/) | github.com/Salman-Khan751

PROFESSIONAL SUMMARY

AI Engineer with a Bachelor's degree in Artificial Intelligence and hands-on experience designing, fine-tuning, and deploying production-grade Machine Learning, NLP, Computer Vision, and Generative AI / Agentic AI systems. Proficient in Large Language Models (LLMs), Transformers, Retrieval-Augmented Generation (RAG), Prompt Engineering, and vector databases using Python, PyTorch, TensorFlow, Hugging Face, LangChain, and LangGraph. Skilled in building and shipping scalable full-stack applications with Flask, FastAPI, React.js, and Node.js, backed by MySQL, MongoDB, Docker, AWS, and CI/CD pipelines. Track record of delivering real-time, low-latency AI solutions across telecom, healthcare, and transportation industries.

EXPERIENCE

Research Associate / AI Engineer

Barani Institute of Information Technology | Jul 2025 – Present

AI Call Intent & Emotion Detection System

Python, Whisper, Wav2Vec2.0, spaCy, Transformers, FastAPI

- Built a telecom call-analysis system that automatically classifies customer intent (upgrade, complaint, cancellation), reducing manual call triage effort for the support team.
- Implemented speech-to-text and emotion detection (angry, neutral, frustrated) by combining Whisper, DistilBERT, and Wav2Vec2.0, enabling real-time sentiment flags on live calls.
- Deployed the pipeline behind a FastAPI service, supporting low-latency inference for production call-center workflows.

Vehicle Detection for Toll Monitoring

Python, YOLOv5, OpenCV, Roboflow

- Developed a YOLOv5-based computer vision system for real-time vehicle detection and classification at toll booths, processing live video streams.
- Improved toll audit accuracy by automatically extracting vehicle counts, vehicle types, and timestamps, reducing reliance on manual logging.

ToxicTrack – Hate Speech Detection System

Python, BERT, spaCy, Transformers

- Designed a real-time NLP pipeline to detect hate speech in user-generated content using transformer-based classification.
- Fine-tuned BERT models to classify toxic, abusive, and threatening language with high accuracy, supporting safer content moderation.

Managed SDLC using Agile practices, Git version control, and collaborative workflows.

SKILLS

Languages: Python, Java, JavaScript, C++

AI & ML: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Agentic AI, LLMs, Transformers, RAG, Prompt Engineering, ANN, RNN, CNN, LSTM, Speech Recognition, Sentiment & Emotion Analysis

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face, spaCy, NLTK, OpenCV, LangChain, LangGraph

Data Science & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Data Annotation, Roboflow

Web & Backend: React.js, Node.js, Flask, FastAPI, REST APIs, GraphQL, HTML5, CSS3, Tailwind CSS

Databases: MySQL, MongoDB, PostgreSQL

DevOps & Tools: Git, GitHub, Docker, AWS, Postman, CI/CD, Agile/Scrum

PROJECTS

MediTranscribe – AI Medical Transcription System (Final Year Project) | 2025-2026

Python, Flask, React.js, MySQL, OpenAI Whisper, BERT, NLP

- Built a full-stack AI medical transcription platform for real-time doctor-patient consultations, integrating live speech-to-text, automated prescription extraction, and EHR data capture.
- Fine-tuned BERT on a custom medical dataset to achieve 95% classification accuracy across clinical text categories (prescriptions, symptoms, diagnostics, CPT codes, notes).
- Engineered secure Flask REST APIs and a React.js frontend with role-based authentication, patient record management, appointment scheduling, and vitals tracking.

NLP Chatbot | 2024

Python, TensorFlow, FastAPI, LLaMA, NLP

- Developed a context-aware, multi-turn conversational chatbot capable of sustaining coherent dialogue across multiple exchanges.
- Fine-tuned a LLaMA-based model on custom datasets and deployed it via a scalable FastAPI service with a web interface.

Movie Recommender System | 2024

Python, Scikit-learn, Pandas, Collaborative Filtering

- Built a personalized recommendation engine using collaborative filtering and similarity-based algorithms to improve content discovery.
- Processed large-scale datasets with Pandas and implemented Scikit-learn pipelines to enhance user engagement and recommendation relevance.

Duplicate Question Detector – Semantic Similarity | 2023

Python, Transformers, spaCy, NLP

- Built a transformer-based semantic similarity model to detect duplicate questions with high accuracy on the Quora Question Pairs dataset.
- Applied NLP feature engineering and text preprocessing with spaCy to deliver a scalable deduplication solution.

EDUCATION

Bachelor of Science in Artificial Intelligence (BSAI) – PMAS Arid Agriculture University, Rawalpindi, Pakistan | 2022 – 2026

CERTIFICATIONS

Cybersecurity – Microsoft / LinkedIn | 2024

Data Analysis – Microsoft / LinkedIn | 2024

Generative AI – Microsoft / LinkedIn | 2023

AWS Lambda – Great Learning | 2023